HOUSTON SPECIAL METALS

SPECIAL METALS, INC.
11625 N HOUSTON ROSSLYN RD
HOUSTON, TX 77086
Phone: 346-374-7900 Fax: 346-374-7899
Website: www.SpecialMetalsInc.com

January 20, 2025 9:46:41AM

REPRINT

Shipper No. 1091437

Bill To CACTUS WELLHEAD LLC
920 MEMORIAL CITY WAY SUITE 300
HOUSTON, TX 77024

APPROVED
By Heather Campbell at 11:47 am, 1/20/25

Ship To CACTUS BOSSIER
4123B MEADOW LANE
BOSSIER CITY, LA 71111

Attn AASHA VALIA 318-752-1616

Customer P.O.#: TRP0048079
Contract:
Sales 1: Robert Husband
Terms: 1% 10 NET 30

Sales Order No: 1091437
F.O.B.: SPECIAL METALS INC
Sales 2: Tommy Westbrook

Due Date: 1/15/25
Ship Via: WILL CALL

QTY	UM	Description	QTY PCS	Length	Weight
		275580			
120.5000	I	HR ALLOY RD 11.000 4340 HT	1	120.5000"	3,312.426
		PN 146814 REV A01 HT 4000015201 CMS-118 REV A CMS-002 REV L CMS-004 REV C CIR-002 REV F CNT-005 REV E Heat: 4000015201 Color Code: YELLOW/WHITE		R4	
					Location: HCY.004
			Totals		**3,312.426**

Shipping Instructions:

Messages:

Carrier _____

Date _____

Pieces _____

Driver Sig _____

Consignee Sig_____

"Please visit our website www.specialmetalsinc.com for our Terms and Conditions"

NUCOR

Mill Certification

03/21/2023

MTR#:1286387-1
Lot #:400001520120
3601 Paul R Lowry Road
Memphis, TN 38109 US
901 786-5900
Fax: 901-786-5901

Sold To: SPECIAL METALS
ATTN ACCOUNTS PAYABLE
2009 S BROADWAY PO BOX 1
MOORE, OK 73160 US

Ship To: SPECIAL METALS
11625 N HOUSTON ROSSLYN RD
HOUSTON, TX 77086 US

Customer PO	463611	Sales Order #	40023674 - 1.1
Product Group	Hot Roll - Engineered Bar	Product #	2309649
Grade	4340	Lot #	400001520120
Size	11"	Heat #	4000015201
BOL #	BOL-1382791	Load #	1286387
Description	Hot Roll - Engineered Bar Round 11" 4340 20' Random 6001-10000 lbs	Customer Part #	274577
Production Date	03/10/2023	Qty Shipped LBS	45280
Product Country Of Origin	United States	Qty Shipped EA	7
Original Item Description		Original Item Number	

I hereby certify that the material described herein has been manufactured in accordance with the specifications and standards listed above and that it satisfies those requirements.

Melt Country of Origin : United States

Melting Date: 02/23/2023

C (%)	Mn (%)	P (%)	S (%)	Si (%)	Cu (%)	Ni (%)	Cr (%)	Mo (%)	V (%)	Al (%)	B (%)
0.41	0.74	0.009	0.009	0.265	0.22	1.75	0.83	0.22	0.005	0.028	0.0003

Sn (%)	Ti (%)	Nb (%)	Ca (%)	Pb (%)	As (%)	Sb (%)	N (PPM)	H (PPM)
0.006	0.002	0.005	0.0002	0.000	0.004	0.001	95	1

DI Calculated (IN) : 7.54

Austenitic fine grain, size 5 or finer, by chemical analysis per the latest revision of ASTM A29.
Reduction Ratio 3.30 : 1

Jominy Actual

J1	J2	J3	J4	J5	J6	J7	J8	J9	J10	J11	J12	J13	J14	J15	J16	J16	J20	J22	J24	J26	J28	J30	J32
60	60	60	60	60	59	59	59	58	59	58	58	58	58	58	58	58	58	58	58	58	58	58	58

ASTM E45 Method A (Worst)

(1) Sulfides T: 1.0 H: 1.0 Alumina T: 1.0 H: 0.0 Silicates T: 0.0 H: 0.0 Globular T: 1.0 H: 0.5

E381 Macroetch

	Macroetch E381 Surface	Macroetch E381 Mid Radius	Macroetch E381 Center
(1)	1	1	1

Other Test Results

ASTM E112 Grain Size : 7

Comments:

AISI/SAE 4340/E4340 HOT ROLLED BAR
MEETS THE REQUIREMENTS OF THE FOLLOWING:
ASTM A29, ASTM A304, ASTM A322 LATEST REVISIONS
UT PER API 6A PSL 3/ASTM A388.

1. Melt/cast practice: electric arc furnace melted, ladle refined, vacuum degassed, continuously cast.
2. Nucor Steel Memphis is compliant with the Pressure Equipment Directive (PED) 97/23/EC 7/2, Annex I, Paragraph 4.3.
3. All manufacturing processes, including melting, have been performed in the U.S.A.
4. No welding or weld repair was performed on this product.
5. No mercury, mercury compounds or mercury containing devices came into contact with this product.
6. Product verified not to exceed background radiation levels.
7. In regards to Section 1502 of the Dodd-Frank Act ("Conflict Minerals"), no covered minerals (tantalum, tin, tungsten or gold) were intentionally added

400- Steven Gage, Chief Metallurgist

NUCOR

Mill Certification

03/21/2023

MTR#:1286387-1
Lot #:400001520120
3601 Paul R Lowry Road
Memphis, TN 38109 US
901 786-5900
Fax: 901-786-5901

to this product.

8. This product conforms to the specifications described on this document. This document may not be reproduced except in full, without written approval of Nucor Corporation.
9. If applicable to the product certified by this report, test procedures are performed in compliance with the current revisions (at the time of testing) of the following ASTM standards:
chemical analysis: E415/E1019, grain size: E112, macrostructure/macroetch: E381, tensile testing: E8/A370, hardness testing: E10 (Type A) /E18/A370. Charpy impact: E23/A370, microcleanliness: E45, decarburization depth: E1077, Jominy: A255.
10. Reported results for ASTM E45, E1077, and E112 are provided as an interpretation of ASTM procedures.
11. If applicable to this report, ASTM E23 tests conducted with 8mm radius striker and 10mm x 10mm V-notch specimen, unless otherwise stated.
12. This document complies with EN 10204 "type 3.1".
13. The following tests are outside the ISO 17025 Laboratory Scope for Nucor Steel Memphis: hydrogen testing and non-destructive testing.
14. Tests not specifically listed in notes 9 and 13, if applicable to this report, are performed by ISO 17025 certified providers outside of Nucor Steel Memphis. Unless otherwise noted, any tests performed by certified providers are within the specific provider's ISO 17025 scope.

400- Steven Gage, Chief Metallurgist

LONE STAR HEAT TREATING CORP.
HOUSTON, TEXAS
SINCE 1945

3939 Blaffer
Houston, Tx 77026
Ph. 713-672-6616
Fax(713)672-9509

Certificate of Heat Treatment And Mechanical Testing

Work Order No: 587469

Cust No: 007544
Cust Name: SPECIAL METALS, INC. Cust PO#: 476375
Quantity: 1 Matl: 4340
Cert Date: 01/10/2025
Description: 1PC@ 11"OD X 126 1/4"LG NEW PART: 275580 HN: 4000015201

NORMALIZE, QUENCH, TEMPER & TEST PER CMS-118 REV A,
TENSILE PER CMS-002 REV L, CVN PER CMS-004 REV C
OK TO SAW CUT 6" TP

Material processed in a Batch Furnace
Controlled by Furnace Instrument Thermocouple Unless Otherwise Noted:

PROCESS	TEMP	TIME	COOLING METHOD
NORMALIZED	1650 F	6.0 HOURS	AIR COOLED
HARDENED	1550 F	7.0 HOURS	OIL 87-106°F
TEMPERED	1100 F	8.5 HOURS	LIQUID COOLED

HARDNESS INSPECTION:
1 PIECE CHECKED IN 2 PLACES 302 HBW / TEST PIECE END CHECKED 302 HBW

MECHANICAL TESTING 17025 ISO/IEC APPROVED CERT# 89036

PROLONGATION
11"OD X6"LG

TENSILE PROPERTIES OF A .500" DIA TB (LONGITUDINAL) MID-RADIUS			
TENSILE	YIELD .2% OFFSET	ELONG in 2"	REDUCTION
145,000 PSI	128,200 PSI	18.8%	58%

CHARPY IMPACTS 10 MM X 10 MM @ MID-RADIUS				
DIRECTION	TEMPERATURE	FT/LBS	MILS LAT EXP	%SHEAR
LONGITUDINAL	0 °F	62-58-58	41-33-31	40-40-40

LONE STAR HEAT TREATING CORP.
HOUSTON, TEXAS
SINCE 1945

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Cert Date: 01/10/2025

SEE ULTRASONIC TEST REPORT ATTACHED

COMMENTS
APPROVED 1100°F TEMPERING TEMPERATURE

Furnaces and instruments are calibrated to AMS 2750G. Reported heat treat times and temperatures are taken from the furnace chart recorder.

Mechanical testing conforms to the requirements of ASTM A370 and only applies to the sample tested or inspected.
Frequency of all testing is dictated by specification and/or customer purchase order.
The report shall not be reproduced except in full without approval of the laboratory.

All work is performed and accepted, unless otherwise agreed to in writing, in accordance with the Lone Star Heat Treating Corporation Master Services Agreement Terms and Conditions which may be found at www.lsht.com and our quality manual QM-01 Rev. 25. While at Lone Star Heat Treating this material did not come in contact with mercury nor was it subject to any weld repairs.

Lone Star Heat TreatingCorp.

Ultrasonic Inspection Report

CACTUS JOB #: | CACTUS PO#: TRP0048079 LN 1 | WO #: 587469 | DATE: 1/13/2025

SUPPLIER NAME: Lone Star Heat Treating Corp | MATERIAL TYPE: 4340

EQUIPMENT DESCRIPTION: NEW PART 275580 | PART NUMBER/REV: 146814 Rev A01

MATERIAL SIZE/THICKNESS: 11"OD x 126 1/4"LG | CUSTOMER PRODUCT IDENTIFIER:

INSPECTION PROCEDURE

NDT SPECIFICATION: CNT-005, Ultrasonic Testing | REVISION: E

Cactus Wellhead NDT Specification contains inspection methodology and acceptance criteria

MODE	☐ SHEAR ☑ LONGITUDINAL	SCANNING METHOD	☑ CONTACT ☑ MANUAL ☐ IMMERSION ☐ AUTOMATIC	PRES. TYPE	☑ A-SCAN ☐ PHASED ARRAY ☐ DIGITAL THICKNESS	AREAS EXAMINED	☑ ALL ACCESSIBLE VOLUME ☐ PRESSURE CONTAINING WELD(S) ☐ CLAD WELD(S) ☐ OTHER (EXPLAIN BELOW)

SURFACE CONDITION: CLEAN, 250 RMS | MATERIAL THICKNESS/DIA: 11"OD

UT INSTRUMENT MAKE/MODEL: Sonatest 250S | S/N AND CAL. DUE DATE: I001449 2/5/2025

COUPLANT: Cellulose | ANGLE: | CABLE TYPE/LENGTH: 6 FT

TRANSDUCER MANUFACTURER: Technisonic | TRANSDUCER SIZE: 1" Dia | TRANSDUCER FREQUENCY: 2.25 MHz

REFERENCE DB: 56 | RECORD ADDITIONAL DB'S FOR: CURVATURE: 2 | ACOUSTIC DIFFERENCES: 0 | SCANNING: (+6) | (TOTAL): 64

REFERENCE STANDARDS

BLOCK SERIAL NUMBER	OVERALL THICKNESS	METAL TRAVEL DISTANCE	REFLECTOR SIZE	REFLECTOR ORIENTATION
LS-4340-1/4-01	3.75"	2.984"	1/4" FBH	Perpendicular
LS-4340-1/4-04	6.75"	5.980"	1/4" FBH	Perpendicular
LS-4340-1/4-07	10.75"	9.996"	1/4" FBH	Perpendicular

INSPECTION RESULTS

WORK ORDER & SERIAL #	HEAT #	HEAT LOT #	SERIAL # (ORIGINAL)	ACCEPT/REJECT QUANTITIES	NCR# (IF APPLICABLE)
587469-1	4000015201			ACCEPT/ 1	

SKETCH AND TECHNIQUE DESCRIPTION (Identify and attach additional drawings/sheets):

MIN. THICKNESS REQUIRED:		MIN. THICKNESS RECORDED:		ACCEPT OR REJECT:	

	Print Name:	Signature:				
INSPECTION PERFORMED BY:	Stephen Lewellen #51312	Stephen Lewellen	Digitally signed by Stephen Lewellen Date: 2025.01.14 14:36:46 -06'00'	LEVEL: III	DATE:	1/13/2025
INSPECTION EVALUATED BY:	Stephen Lewellen #51312	Stephen Lewellen	Digitally signed by Stephen Lewellen Date: 2025.01.14 14:37:06 -06'00'	LEVEL: III	DATE:	1/13/2025
CUSTOMER REPRESENTATIVE:					DATE:	

Signature of the NDE Technician substantiates training to the NDT Specification and attests to inspections performed in accordance with the stated inspection methodology. Supplier NDE Technician qualifications and current eye examinations are provided and maintained by the company.